

Gradient Starvation in Low-Rank Language Model Pretraining

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Abstract

Low-rank (LR) pretraining underperforms full-rank (FR) pretraining, but why the gap persists remains unclear. Using GPT-2 on BabyLM, we show the representational split between them is not fixed at initialisation: Spectrum-matched initialisation eliminates the representational gap at step zero, but it is reasserted within a few training steps, implicating training dynamics over initialisation. We further show gradient dimensionality is a causally separable driver of this split, not merely a restatement of the weight factorization. Restricting or expanding gradient dimensionality of LR models directly shifts representational alignment with the FR counterpart independent of weight rank. Finally, optimisation success and representational alignment dissociate, that is, spectral normalisation and other interventions improve perplexity without recovering FR–LR similarity. Also, we observed that representational alignment predicts downstream competence better than perplexity does. These results support gradient starvation as a genuine partial mechanism and suggest that efficient pretraining should be evaluated by both representational trajectory and performance, not solely the latter.

1 Introduction

Efficient language model pretraining increasingly relies on low-dimensional training constraints. Low-rank parametrisations reduce memory and compute by restricting weight matrices to a factorised form $\mathbf{W} = \mathbf{B}\mathbf{A}$ with rank $r \ll \min(m, n)$ (Hu et al., 2022; Lialin et al., 2024). Gradient projection methods compress optimiser states while preserving full-rank weights (Zhao et al., 2024), while structured accumulation schemes recover expressivity through periodic restarts (Lialin et al., 2024; Li et al., 2025). Together these approaches have made low-rank pretraining a practical choice for resource-constrained settings (Warstadt et al., 2023a; Hu et al., 2024; Wei et al., 2024). Yet the

performance gap with full-rank (FR) pretraining persists across methods and scales (Lialin et al., 2024; Zhao et al., 2024; Diehl Martinez et al., 2024). The field’s response has been almost entirely engineering: Design a better method that narrows the gap. What has received far less attention is the mechanistic question: *What does the rank constraint do to the representations a language model learns, and why does the damage resist correction?*

This paper provides evidence for a mechanistic account of low-rank (LR) pretraining dynamics in language models. Using GPT-2 pretrained on the BabyLM Strict-Small corpus (Hu et al., 2024) as our primary setting, we conduct a systematic analysis organised around six objectives spanning divergence timing, stabilisation mechanism, gradient dimensionality, spectral causality, performance scaling, and recoverability. We use Centered Kernel Alignment (CKA; Kornblith et al., 2019) to track representational divergence, effective rank (Roy and Vetterli, 2007) to characterise gradient dimensionality, and targeted spectral interventions, including direct manipulation of gradient dimensionality independent of rank, to establish causality. We validate that this representational divergence carries functional consequences via a downstream probing task, and confirm that CKA measurements have mutual alignment with k-nearest-neighbours, an alternative representational metric. We complement language findings with a parallel ViT (Dosovitskiy et al., 2021) analysis on CIFAR-10 as a supplemental cross-domain check.

The split is not fixed by initialisation, and reasserts during training. At 8,192 tokens, approximately one batch, LR models at all ranks are already separated from FR in representation space. This split persists even under a spectrum-matched initialisation that eliminates it at true step zero. The advantage does not survive training. The split reappears within a single step and the model ultimately

diverges from FR more than under standard initialisation. Initialisation is therefore not the origin of the split; the rank constraint sustains it continuously during training.

Gradient starvation is the sustaining mechanism.

We identify a mechanism that explains much of this gap, which we call gradient effective rank starvation. FR’s gradient effective rank grows from 306 to 510 over training, while LR’s remains bounded, and unlike vision, where the ratio is constant, the gap in language *widens*. We show this constraint is causally separable from the weight factorisation. Restricting gradient dimensionality directly, with weights remaining full-rank reproduces the divergence. Similarly, periodically refreshing the update subspace under a fixed weight rank recovers alignment without changing the rank itself. We conclude that gradient dimensionality, not the factorisation per se, is the operative constraint.

Stabilisation reveals a language-specific pathology.

All LR language model runs stabilise by *exhaustion*, i.e., representations drift far from initialisation before step-to-step changes become small, indicating the model has run out of productive gradient directions. FR language models also stabilise by exhaustion, which is unlike FR vision models, whose shallow layers show clean convergence. Two independently trained FR language models *diverge* in representation space over training rather than converging. The rank constraint amplifies a pathology already latent in language pretraining.

Spectral geometry is causal at initialisation, but its effect does not persist.

Intervening at a strategic chosen timepoint t^* to flatten the LR model’s singular value spectrum raises CKA alignment by 288% and reduces perplexity by 61%, confirming that spectral geometry causally drives the early split. However, this gain does not sustain. Gradient flow overwrites the correction within a few updates, and the same dissociation appears with hyperparameter-tuned LR, which reaches a much lower perplexity of 138 yet shows almost no gain in representational alignment. Performance improves without closing the representational gap, regardless of the lever used.

The gap is only partially recoverable. Full fine-tuning of a trained LR model closes only 12.5–42% of the performance gap in language depending on training budget; reinitialising all transformer blocks

recovers up to 73%, yet a substantial residual persists. Restricting gradient dimensionality alone, with no weight factorisation, closes at most half of LR’s representational gap to FR. Neither post-hoc adaptation nor full-ranked weights fully escapes the distinct representation basin of FR and LR shown in prior work (Lyle et al., 2023; Dohare et al., 2024).

Contributions. We demonstrate that:

1. The FR–LR representational split is not fixed by initialisation. It reasserts within the first few training steps regardless of starting point, implicating the rank constraint acting during training.
2. Gradient effective rank starvation contributes to the gap: a widening deficit in gradient dimensionality, more severe and dynamic in language than in vision.
3. Gradient dimensionality is causally separable from the weight factorisation: restricting or expanding it directly, independent of rank, shifts representational alignment accordingly.
4. Spectral correction, hyperparameter tuning, and post-hoc fine-tuning each improve performance without closing the representational gap (Roy and Vetterli, 2007; Kornblith et al., 2019; Lyle et al., 2023; Dohare et al., 2024).
5. Representational alignment predicts downstream grammatical competence better than perplexity does, confirmed under an independent, non-CKA alignment metric.
6. Cross-domain validation on ViT/CIFAR-10 (Dosovitskiy et al., 2021) shows core phenomena replicate while revealing systematic domain differences.

All code, training scripts, probing pipelines, and collated results are released for reproducibility.¹

2 Background

Low-rank pretraining constrains each weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ to a factorised form $\mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{B} \in \mathbb{R}^{m \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times n}$, and $r \ll \min(m, n)$, reducing trainable parameters from mn to $r(m + n)$ (Lialin et al., 2024; Zhao et al., 2024). The performance cost of this constraint is well documented; LR pretraining consistently

¹https://github.com/ovasoft/rank_dynamics

underperforms FR pretraining (Lialin et al., 2024; Zhao et al., 2024; Wei et al., 2024).

Prior work offers partial clues about why this occurs. Diehl Martinez et al. (2024) show that layers with lower effective rank in small language models exhibit slower and less stable convergence during pretraining. Godey et al. (2024) establish a theoretical lower bound on achievable loss when model width is insufficient for the rank of the target output distribution. However, neither work characterises *when* the representational divergence between FR and LR arises, what sustains it during training, or whether it is reversible. Intuitively, if FR training can explore many independent update directions while LR training cannot, the two models may diverge functionally even under identical optimisation settings. This paper provides evidence for a mechanistic account of that divergence.

Our focus differs from prior work. Where previous work asks how to close the performance gap, we ask what creates and sustains it at the representational and optimisation level. This distinction motivates the mechanistic tools introduced below.

2.1 Metrics

Centered Kernel Alignment (CKA). We measure representational divergence using linear CKA (Kornblith et al., 2019), which is invariant to orthogonal transformations and isotropic scaling and therefore well-suited to comparing representations across independently trained models without a priori alignment. For activation matrices $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{Y} \in \mathbb{R}^{n \times q}$ collected over n examples:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (1)$$

We report three variants: *cross-run CKA* between an LR model and a fixed FR reference (FR–LR divergence); *self-CKA* between consecutive checkpoints of the same run (representational velocity); and *init-CKA* between a checkpoint and its own initialisation (cumulative drift). All CKA values are means across transformer layers unless stated otherwise; layer-wise analyses showing consistent trends are provided in Appendix G.

Gradient effective rank. We characterise gradient dimensionality using effective rank (Roy and Vetterli, 2007), the exponential Shannon entropy of the normalised singular value spectrum of gradient matrix $\mathbf{G}$:

$$\text{eRank}(\mathbf{G}) = \exp\left(-\sum_i p_i \log p_i\right), \quad (2)$$

$$p_i = \frac{\sigma_i(\mathbf{G})}{\sum_j \sigma_j(\mathbf{G})}. \quad (3)$$

eRank measures how broadly gradient energy is distributed across singular directions. Higher effective rank implies that gradient energy is distributed across more independent update directions, increasing the dimensionality of the optimisation search space available at each step. We report the ratio $\text{eRank}(G_{\text{FR}})/\text{eRank}(G_{\text{LR}})$ to quantify FR’s relative gradient richness.

Mutual k -nearest-neighbour alignment. As a representational alignment metric independent of CKA, we adopt mutual k -NN alignment (Huh et al., 2024). For a shared set of inputs, each point’s k nearest neighbours are identified separately in each model’s representation space, using cosine similarity. Alignment is the mean fraction of neighbours shared between the two spaces, in $[0, 1]$, with 1 indicating identical local neighbourhood structure and $k/(n-1)$ the expected value under no relationship. Unlike CKA, which measures global linear-subspace correspondence, mutual k -NN alignment is sensitive only to local neighbourhood agreement, providing an independent check on our CKA-based findings.

Gradient starvation. We define *gradient starvation* as the persistent restriction of gradient updates to a low-dimensional update subspace relative to FR training. Operationally, we identify starvation when the ratio $\text{eRank}(G_{\text{FR}})/\text{eRank}(G_{\text{LR}})$ remains substantially above 1 throughout training, indicating that the optimiser cannot compensate for the rank constraint regardless of how many steps are taken. Under starvation, entire update directions available to FR training are absent from the LR update subspace.

Stabilisation mechanism. A run stabilises at checkpoint t_s when self-CKA first exceeds $\tau = 0.995$ and remains above it, indicating that representations are no longer changing meaningfully between updates. We classify the stabilisation mechanism using init-CKA at t_s :

- **Convergence** ($\text{init-CKA}(t_s) \geq 0.65$): representations remain close to initialisation at stabilisation, consistent with a solution found early in the optimisation trajectory.
- **Exhaustion** ($\text{init-CKA}(t_s) < 0.65$): representations have undergone substantial drift

from initialisation before stabilising, consistent with the model having explored available update directions without reaching a high-quality solution.

The threshold $\tau_{\text{init}} = 0.65$ is motivated empirically and its sensitivity is discussed in Appendix F.

3 Experimental Setup

3.1 Models and Domains

Language (primary). We pretrain GPT-2 (Radford et al., 2019) on the BabyLM Strict-Small corpus (Hu et al., 2024): 12 transformer layers, 12 attention heads, embedding dimensionality 768, context length 128. Training uses a 50M-token budget (five passes) with AdamW (Loshchilov and Hutter, 2019), learning rate 10^{-4} , batch size 64, and cosine decay. Checkpoints are saved at {8K, 32K, 131K, 524K, 1M, 2M, 5M, 10M, 20M, 30M, 40M, 50M} tokens. We evaluate on held-out validation perplexity. All experiments use BF16 precision on NVIDIA A100 GPUs; full hyperparameter tables are provided in Appendix H.

Vision (cross-domain check). Our ViT (Dosovitskiy et al., 2021) is pretrained (patch size 4, embedding dimensionality 192, depth 9, heads 3) on CIFAR-10 for 20 epochs with AdamW (learning rate 10^{-3} , cosine decay), saving one checkpoint per epoch. We evaluate on top-1 validation accuracy.

Why BabyLM. The small-scale controlled setting is a deliberate choice. Dense checkpointing is practical, rank sweeps across six values are feasible without distributed-training confounds, and the fixed data budget isolates rank as the causal variable. We hypothesise that the observed mechanisms are not unique to small-scale settings, though validating this at frontier scale remains future work. We do not claim the specific performance numbers generalise; we claim the mechanistic account motivates hypotheses testable at larger scale.

3.2 Low-Rank Parametrisation

For each LR run, all linear projection matrices (query, key, value, output, and both feed-forward projections) are replaced with factorised matrices $\mathbf{W} = \mathbf{B}\mathbf{A}$ at rank r . We sweep $r \in \{8, 16, 32, 64, 128\}$ for vision and $r \in \{8, 16, 32, 64, 128, 256\}$ for language, alongside a full-rank (FR) baseline. All hyperparameters are held constant across FR and LR runs.

By default, $\mathbf{B}$ and $\mathbf{A}$ are initialised independently at random. As a control, we also construct a spectrum-matched initialisation, in which $\mathbf{B}$ and $\mathbf{A}$ are set to reconstruct the best rank- r approximation of FR’s own initial weight, obtained via truncated SVD. This matches LR’s starting spectrum to FR’s as closely as any rank- r factorisation can, and isolates whether the initial representational split reflects the rank constraint itself or an artefact of the default initialiser.

Statistical reliability. We train two independently seeded runs per condition; qualitative trends are consistent across seeds, and values are reported as means (Appendix I).

3.3 Probing Pipeline

After training, each checkpoint is probed with a fixed held-out batch of 512 examples drawn from the validation set and held constant across all checkpoints and models, ensuring all measurements are directly comparable. For language, activations are the hidden state of the last non-padding token; for vision, the [CLS] token. Layer-wise activations are collected from each transformer block before the residual addition. Cross-run CKA is computed against the corresponding layer of the same fixed FR reference model using the same held-out batch. Mutual k -NN alignment ($k = 10$) and BLiMP (Warstadt et al., 2023b) accuracy over 1,600 grammatical minimal pairs are computed similarly, as an independent alignment metric and a functional probe, respectively (full methodology provided in Appendix J, layer-wise plots corroborating mean-CKA trends in Appendix G).

Gradient eRank is estimated via a separate forward-backward pass on the same held-out batch. For each weight matrix, the gradient is treated in its native 2D matrix shape; higher-order tensors (e.g., convolutional kernels) are reshaped to their corresponding matrix form before SVD. Economy SVD is applied and eRank computed via Equation (2); reported values are averaged across transformer layers.

3.4 Interventions

Spectral correction at t^* . At the first saved checkpoint t^* , we apply *alpha-flattening*: the singular values of each weight matrix in the LR model are rescaled to a uniform distribution — a high-entropy singular value distribution resembling the comparatively isotropic spectra of randomly ini-

tialised FR weights — while singular vectors are left unchanged. This provides intervention evidence for the role of spectral geometry while controlling for other accumulated training differences. We additionally test a *drift-scaled* variant (each matrix scaled by its measured init-CKA drift) and a *combined* variant (both corrections applied simultaneously).

Dynamic spectral normalisation (DSN). To test whether sustained spectral correction can maintain alignment during training, we apply three DSN modes every 1,000 steps, immediately before the optimiser update. *Static*: singular values are rescaled to a fixed uniform target, with Frobenius norm preserved. *Dynamic*: singular values are rescaled to match the distribution of the temporally matched FR checkpoint, obtained from a parallel FR run saved at identical intervals. *Flat*: a uniform singular value distribution is enforced at every step, preserving Frobenius norm. All modes operate on weight matrices directly and do not modify optimiser states.

Layer reinitialisation (k -sweep). To assess post-hoc recoverability, we reinitialise the first k transformer blocks of a trained LR model to full-rank random weights and fine-tune, sweeping $k \in \{0, 1, 2, 3, 4, K\}$ where K denotes all blocks. Each condition is evaluated under a *short* budget (matched to original training length) and a *convergence* budget (run until validation metric plateaus).

Gradient truncation (GradTrunc) To isolate gradient dimensionality from weight rank, we train a full-rank model whose gradient is projected onto a fixed rank-8 random orthonormal subspace at every step, chosen once at initialisation and never updated. Target weights are excluded from weight decay to avoid full-rank drift the projection does not constrain.

ReLoRA To test whether periodically expanding the update subspace restores alignment under a fixed weight rank, we train a rank-8 adapter merged into the base weight and reinitialised every 5M tokens, with optimiser state reset at each merge.

Rank-tuned hyperparameters (LR-tuned) To rule out FR-favourable hyperparameter confounds, we sweep learning rate ($0.5\text{--}3\times$) and initialisation scale ($0.5\text{--}2\times$) for LR at $r = 8$ under a reduced budget, then retrain the best configuration ($3\times$ LR, $0.5\times$ scale) at full budget.

3.5 Normalised Performance Gap

We report separate gap measures for each metric direction. For accuracy (higher is better):

$$\delta_r^{\text{acc}} = \frac{m_{\text{FR}} - m_{\text{LR}}}{m_{\text{FR}}}, \quad (4)$$

and for perplexity (lower is better):

$$\delta_r^{\text{ppl}} = \frac{m_{\text{LR}} - m_{\text{FR}}}{m_{\text{FR}}}. \quad (5)$$

In both cases, $\delta_r > 0$ indicates LR underperforms FR and $\delta_r = 0$ indicates parity, making comparisons across metrics interpretable without directional ambiguity.

4 Results

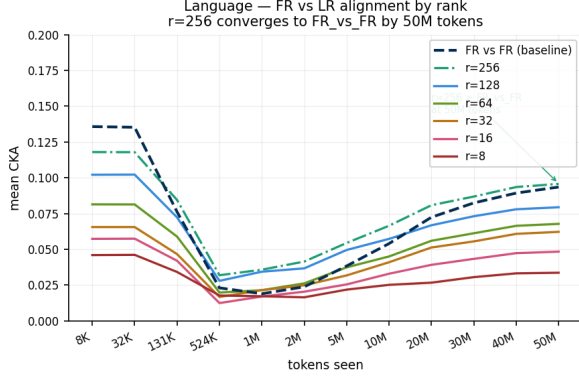
We organise results around six aims. Language findings are reported first throughout; vision findings serve as cross-domain comparisons. Table 1 summarises headline numbers; seed-level variance is reported in Appendix I and is small relative to the effects described here.

Core result. At 8,192 tokens — approximately one batch — LR models at all ranks are already separated from FR in representation space. For LR_r8², cross-run CKA against FR is 0.046 ± 0.003 , a 66% reduction relative to the FR–FR baseline (0.136 ± 0.008). The gap changes by less than 0.001 between the first and second checkpoints, indicating the split is established prior to substantial optimisation rather than emerging gradually (Figure 1a). Vision supports this: At epoch 0, before any gradient update, FR–LR CKA for LR_r8 (0.084 ± 0.005) is already far below the FR–FR baseline (0.736 ± 0.012).

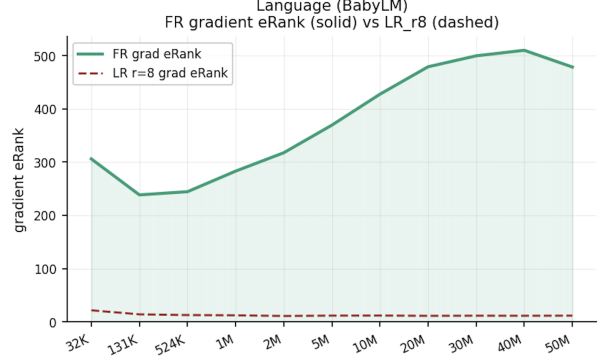
Supporting result: rank ordering. Higher rank produces less divergence from FR at every checkpoint in both domains, with monotonic rank ordering stable throughout training. In this setting, LR_r256 reaches near-parity with FR–FR by 50M tokens, while LR_r128 remains below. Exact values are given in Table 1.

Domain difference. Normalising FR–FR CKA to $t=0$, vision FR seeds converge over training ($1.07\times$) while language FR seeds diverge ($0.69\times$) — two independently trained FR models become less similar in language but more similar in vision. This domain-level difference motivates the

²Throughout the paper, we use the suffix *_rn* to denote rank $r = n$.



(a) Cross-run CKA between FR and LR models across all token checkpoints (BabyLM Strict-Small). Dashed line: FR–FR baseline. All LR ranks fall below FR–FR at 8K tokens; the gap changes by < 0.001 between the first two checkpoints, making a transient initialisation-only effect unlikely. FR–FR CKA decreases over training (language FR seeds diverge; vision FR seeds converge — Appendix A).



(b) Absolute gradient eRank of FR (solid) and LR_r8 (dashed) over training. In language, FR gradient eRank expands from 306 to 510, while LR_r8 remains near its structural ceiling. The growing gap in language underlies the escalating gradient starvation reported in Figure 2 and Appendix C.

Figure 1: (a) The FR–LR split is established before meaningful learning (Aim 1). (b) FR gradient dimensionality expands in language while LR’s remains bounded (Aim 3). Together these panels motivate the two-phase mechanistic account: The split exists structurally from initialisation and gradient starvation widens it throughout training. Full eRank ratio trajectories are given in Appendix C.

investigation of whether the split is associated with persistent optimisation differences, addressed in §4.2.

4.1 Aim 2: Language Pretraining Stabilises by Exhaustion

Observation. Under our operational definition (§2), every language run, FR and all LR ranks, stabilises in the exhaustion regime. Mean init-CKA at stabilisation ranges from 0.44 (FR) to 0.67 (LR_r256), all below the 0.65 convergence threshold. In vision, the mechanism is rank-dependent: LR_r128 stabilises by convergence (mean init-CKA = 0.68, 100% of layers); LR_r8 and LR_r16 stabilise by exhaustion (≈ 0.13 , 100% of layers).

Interpretation. The exhaustion classification for the FR language model itself, not only for LR runs supports the view that exhaustion-type stabilisation is a property of language pretraining at this scale. Higher rank stabilises earlier in both domains (LR_r256 at ~ 15 – 20 M tokens; LR_r8 at ~ 40 M tokens in language), which accords with the interpretation that higher-rank models traverse productive gradient directions more rapidly. Alternative explanations are discussed in Appendix F. Together, these observations raise the question of whether LR and FR models differ systematically in gradient dimensionality throughout training, addressed next.

4.2 Aim 3: Evidence for Gradient Starvation

Observation. Figure 2 shows the gradient eRank ratio $\text{eRank}(G_{\text{FR}})/\text{eRank}(G_{\text{LR}})$ across training. In language, the ratio grows substantially throughout training at all ranks: for LR_r8 it rises from 14.3 to 42.3 over the full training run, driven by FR’s gradient eRank expanding from 306 to 510, while LR’s remains near its structural ceiling. In vision, the ratio is stable across all epochs (CV 2–7% for $r \leq 64$), established at the first update and not closing. Doubling rank approximately halves the final ratio in both domains; exact values are reported in Table 1.

Interpretation. These observations are consistent with gradient starvation as defined in §2: LR models operate in a persistently reduced update subspace relative to FR, and in language this deficit widens as training progresses. The later into training, the further LR falls behind FR’s gradient richness; an escalating constraint rather than a fixed one. We establish that this constraint is causally separable from the weight factorisation itself in the subsequent paragraph. This motivates testing whether modifying spectral geometry at initialisation can counteract the divergence, addressed in §4.3.

Direct gradient-dimensionality intervention.

Observation. To test whether gradient starvation

Aim 3 — O3.1: Is the gradient eRank gap constant?

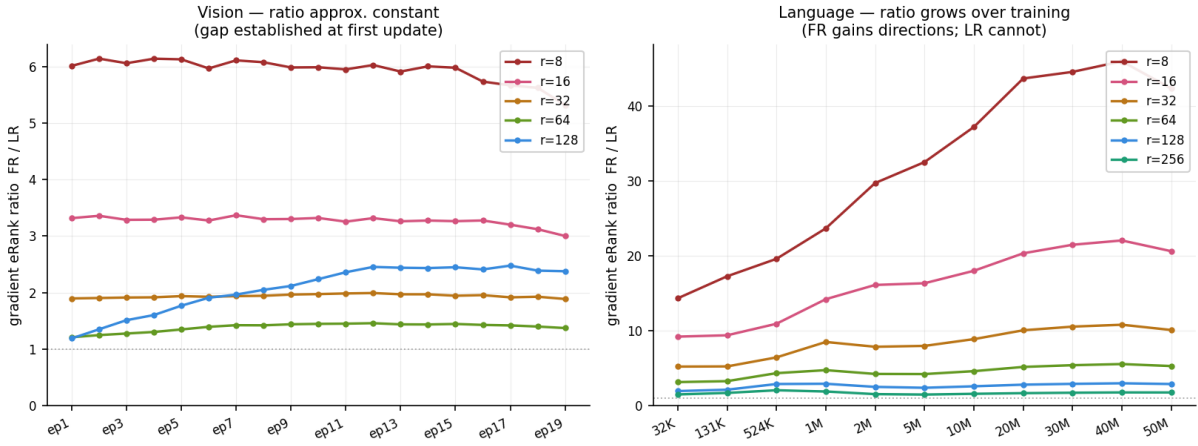


Figure 2: **Gradient starvation sustains and widens the gap throughout training (Aim 3).** Gradient eRank ratio $\text{eRank}(G_{\text{FR}})/\text{eRank}(G_{\text{LR}})$ over training. *Right* (language): The ratio grows substantially throughout training as FR continues to expand its effective gradient dimensionality while LR’s remains bounded by the rank constraint. Gradient starvation is an escalating constraint in language. *Left* (vision): The ratio is stable across all epochs (CV 2–7% for $r \leq 64$), established at the first update and not closing thereafter. Gradient starvation is a fixed constraint in vision. Supporting absolute eRank and rank-scaling analyses in Appendix C.

is causally separable from the weight factorisation itself, rather than a restatement of it, we intervene on gradient dimensionality directly. GradTrunc restricts gradients to a fixed rank-8 subspace on an otherwise full-rank model; ReLoRA holds LR_r8’s weight rank fixed but periodically refreshes the update subspace. Both interventions shift both gradient eRank and CKA away from LR_r8’s baseline in the predicted direction, that is, GradTrunc reaches a final gradient eRank of 77.6 and closes 50% of LR_r8’s CKA gap to FR; ReLoRA reaches 60.1 and closes 70%.

Interpretation. Since GradTrunc’s weights are never factorised, and ReLoRA holds weight rank constant while varying only the update subspace, neither result can be explained by the factorisation alone. This directly addresses the correlated-factors caveat above: gradient dimensionality, not the rank constraint per se, is doing causal work.

4.3 Aim 4: Spectral Geometry Drives the Early Split

O4.1 — Intervention at t^* . Alpha-flattening at t^* raises cross-run CKA from 0.082 to 0.318 (+288%) and reduces validation perplexity from 490 to 188 (−62%) in language. Drift-scaling alone provides minimal CKA improvement. The combined variant underperforms alpha-flat alone, supporting the view that spectral geometry rather than drift magnitude is the operative factor. Vision

results are consistent: CKA rises by 62% and accuracy improves by 13.5 pp (Table 2). These results provide intervention evidence for the role of spectral geometry at initialisation in driving the early split.

O4.2 — DSN trajectories. *Observation.* Despite perplexity reductions of 75–77% relative to untreated LR (DSN-static: ppl 117; DSN-dynamic: 115; DSN-flat: 107; LR baseline: 457), no DSN mode improves cross-run CKA above the untreated LR baseline at the end of training. Final CKA values cluster near the untreated LR baseline (≈ 0.033 – 0.036) across all three DSN modes.

Interpretation. Task performance and representational alignment are dissociated under DSN: the intervention substantially improves learning outcomes without producing more FR-like representations. This supports the view that ongoing gradient starvation prevents sustained alignment regardless of spectral initialisation. Crucially, this dissociation — improved perplexity without improved FR-alignment — demonstrates that optimisation success and representational similarity are separable phenomena, a distinction we return to in §5.

O4.3 — Spectrum-matched initialisation and true step-zero. *Observation.* To test whether the split reflects spectral geometry at initialisation or the rank constraint itself, we initialise LR_r8 with factors set to FR’s own best rank-8 approximation

Model	CKA to FR	BLiMP	PPL	eRank growth	Self-CKA
FR	0.9800	0.6338	46.98	$5.98\times$	1.0000
LR_r8	0.0349	0.5563	456.60	$0.65\times$	0.7050
GradTrunc8	0.0645	0.5775	72.81	$2.48\times$	0.8499
ReLoRA_r8	0.0527	0.4950	604.58	$1.91\times$	0.8839
LR_specmatch8	0.0287	0.5325	438.75	$4.73\times$	0.6758
LR_r8_tuned	0.0398	0.5200	138.24	$0.62\times$	0.7300
LR_r8_dsn_static	0.0313	0.5150	117.18	$0.93\times$	0.9507
LR_r8_dsn_dynamic	0.0400	0.5144	114.73	$0.94\times$	0.9556
LR_r8_dsn_flat	0.0328	0.4875	109.67	$0.83\times$	0.9496

Table 1: Raw results across all evaluated models. eRank growth is the ratio of final to initial mean gradient effective rank (attention layers); self-CKA is the mean within-run CKA between consecutive checkpoints.

at true step zero. This eliminates the split entirely: CKA reaches 123.6% of the natural FR–FR baseline, matching or exceeding the variability between two independently-initialised FR models. The advantage does not survive training. By the first checkpoint, CKA has fallen 78% from its step-zero value and already sits below default-initialisation LR_r8’s own CKA at that point. By the final checkpoint, spectrum-matched LR_r8 reaches only 30.7% of the FR–FR baseline, *below* the 37.2% attained under default initialisation.

Interpretation. A favourable, FR-matched starting point confers no lasting protection: the split re-opens within a single training step and the spectrum-matched model ultimately diverges from FR *more* than default initialisation does. This rules out initialisation as the origin of the split. The persistent rank-8 gradient constraint during training, not the starting point, is responsible.

4.4 Aim 5: Performance Gap Mirrors Gradient Starvation

The performance gap scaling (Appendix E, Figure 10) closely mirrors the gradient eRank ratio scaling in §4.2: doubling rank approximately halves both the eRank ratio and the normalised performance gap δ_r . In language, δ_r^{ppl} decreases monotonically from 8.72 ($r=8$) to 0.28 ($r=256$), with LR_r256 approaching FR at a level consistent with its near-parity CKA in §4. In vision, δ_r^{acc} decreases from 0.59 ($r=8$) to 0.24 ($r=64$) but shows a non-monotonic anomaly at $r=128$ ($\delta = 0.31$), consistent with overfitting at higher capacity on CIFAR-10 and absent in language. Language gaps are $\sim 15\times$ larger than vision at equivalent ranks, which aligns with the proportionally larger language eRank ratios in §4.2 and with the theoretical lower bound of Godey et al. (2024). Full results are given in Appendix E.

Rank alone is not the only lever on this gap: tuning learning rate and initialisation scale for LR_r8 reduces perplexity by 69.7% (from 457 to 138), yet closes only $\sim 8\%$ of its CKA gap to FR and leaves gradient eRank essentially unchanged. Performance and the mechanisms we attribute it to move largely independently under hyperparameter tuning, mirroring the dissociation observed under DSN in §4.3.

4.5 Aim 6: The Gap is Only Partially Recoverable

Full fine-tuning. Full fine-tuning of a trained LR_r8 model closes 10.7% of the vision gap and 42.0% of the language gap under a convergence budget. Both domains leave substantial gaps unrecovered despite extended optimisation (Table 1 and Appendix E).

Layer reinitialisation. Figure 11 shows recovery as a function of k reinitialised blocks. In *language*, recovery increases monotonically: $k=2$ closes 31.2%; $k=4$ closes 54.6%; all blocks ($k=11$) closes 73.0%. In *vision*, recovery is non-monotonic: $k=1$ closes 17.2% but $k=2$ closes only 8.7%, with substantial recovery appearing only at $k=9$ (all blocks, 49.6%). This all-or-nothing profile in vision, compared with the smooth monotonic profile in language, mirrors the Aim 1 finding that CKA profiles are layer-uniform in language but layer-graded in vision.

Residual gap. Under the most aggressive recovery condition, 51% of the vision gap and 58% of the language gap remain unrecovered. The interpretation of this persistent residual is discussed in §5.

4.6 Representational Alignment Predicts Function Better Than Perplexity

Across eight interventions, CKA correlates with BLiMP accuracy considerably more than perplexity does ($r=0.40$ vs. $r=-0.08$). DSN is the clearest case: It improves perplexity by 74–77% over untreated LR_r8 yet remains at chance on BLiMP across all three variants, while untreated LR_r8, despite far worse perplexity, stays significantly above chance. Mutual k -NN alignment (Appendix K.2) confirms the representational gap is not CKA-specific: GradTrunc, ReLoRA, and DSN-static all remain well below the natural FR–FR baseline under this independent metric. Full per-run results, significance tests, and the mutual k -NN trajectories are reported in Appendix K.

5 Discussion

Our results support a two-phase mechanistic interpretation of low-rank pretraining. The split is present before meaningful learning: at 8K tokens, all LR ranks are already below the FR–FR baseline. This is not a fixed consequence of initialisation. A spectrum-matched initialisation eliminates the split entirely at true step zero, yet the split reasserts within a single training step and the model ultimately diverges from FR more than under standard initialisation. Initialisation therefore does not explain the split; a persistent gradient constraint acting throughout training does. This constraint is causally separable from the weight factorisation. Restricting gradient dimensionality alone, with weights left full-rank, reproduces divergence from FR, closing up to half of LR’s gap. The remaining half is attributable to the factorised parametrisation itself, which we leave to future work. Spectral correction and gradient dimensionality each contribute to representational recovery, however, neither is independently sufficient for the recovery.

A second finding is that optimisation success and representational similarity to FR are separable. DSN achieves substantially better perplexity than untreated LR while final CKA remains unchanged, and CKA is uncorrelated with perplexity across every intervention tested. We treat the two as complementary, not substitutable, metrics, each exposing failure modes the other misses. This matters functionally; representational alignment predicts downstream grammatical competence considerably better than perplexity does, and DSN’s representations

remain at chance on this functional probe despite its perplexity gains (§4.6). Existing performance-based evaluations of low-rank pretraining may systematically underestimate its representational cost. A model can achieve low perplexity while occupying a representational organisation substantially different from, and less functionally competent than, a full-rank model, relevant for transfer learning. Closing the performance gap and closing the representational gap are distinct targets.

We observe systematic differences between the language and vision modalities (Appendices A–C); our findings are strongest as characterisations of language-specific dynamics, and whether they hold at frontier scale or across other architectures is left to future work.

6 Conclusion

We presented a mechanistic analysis of low-rank pretraining across six objectives. The FR–LR split is not fixed by initialisation: a spectrum-matched initialisation eliminates it at true step zero, yet it reasserts within a single training step, implicating a persistent gradient constraint rather than the starting point. This constraint is causally separable from the weight factorisation; restricting or expanding gradient dimensionality directly shifts alignment with FR, independent of weight rank, though a substantial residual gap remains unexplained. A key finding is the dissociation between optimisation success and representational alignment: Spectral normalisation reduces perplexity by 75–77% while final CKA is unchanged, and alignment predicts downstream competence better than perplexity does, confirmed under an independent, non-CKA metric. Efficient pretraining methods should hence be evaluated not only by performance, but also by the representational and optimisation trajectories, and their functional consequences, through which performance emerges. Code and results are released.³

7 Limitations

Scale. All experiments use small models pre-trained on a 10M-word corpus. GPT-2 on BabyLM has 85M parameters; our ViT has 4.3M. We hypothesise that the mechanisms we identify are not unique to this scale, but we have not run a larger-scale validation to confirm this. Whether the effects

³https://github.com/ovasoft/rank_dynamics

we report hold at frontier scale remains open and is left to future work.

Single optimisation method. All experiments use AdamW. The gradient starvation finding may be optimiser-dependent: methods that explicitly expand the gradient subspace, such as second-order optimisers or random gradient projection, might partially counteract the rank constraint. Whether the mechanisms we identify hold under other optimisation techniques is an open question.

Single architecture family. We study standard transformer architectures with dense attention and feed-forward layers. The mechanisms may differ for architectures with different inductive biases — state-space models, mixture-of-experts networks, or linear attention variants — where the role of weight matrix rank is structurally different.

Metrics as proxies. Effective rank is a proxy for gradient dimensionality, not a direct measure of optimisation freedom. CKA measures representational geometry, and we test whether this geometry has functional consequences in §4.6: CKA predicts downstream grammatical competence considerably better than perplexity does ($r=0.40$ vs. $r=-0.08$). With only eight interventions, this correlation is suggestive rather than decisive evidence that CKA is a reliable proxy for function. Mutual k -NN alignment, used as an independent check, does not always agree with CKA on the relative ordering of conditions: it ranks DSN-static above untreated LR_r8, the reverse of CKA’s ordering. We report both metrics rather than treating either as definitive, and note that other analyses — neural tangent kernel comparisons, probing accuracy, or additional functional similarity metrics — could reveal further aspects of the FR–LR divide not captured here.

Two seeds. Due to the computational cost of full checkpointed rank sweeps, our core rank-sweep results use two seeds per condition. This is sufficient to confirm qualitative stability of the large-effect-size trends we report, but does not support precise confidence intervals. Our causal-intervention results — GradTrunc, ReLoRA, spectrum-matched initialisation, and hyperparameter-tuned LR — use a single seed each. Findings that are sensitive to seed, including these interventions and the exact stabilisation checkpoint, should be interpreted with appropriate caution.

Causal interpretation. The spectral intervention at t^* provides evidence for the causal role of spectral geometry, but the intervention modifies weight matrices in ways that could affect properties beyond the singular value spectrum, including the alignment of singular vectors across layers. We interpret the results as intervention evidence rather than proof of causation, and note that disentangling the contributions of spectral shape, singular vector geometry, and weight norm remains an open problem.

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A Aim 1: Supporting Divergence Figures

This section provides the vision replication of the pre-existing split finding and the normalised divergence analyses that support the rank-ordering claim in §4. Figure 3 mirrors Figure 1(a) for vision, confirming that the split is present before any

gradient update and that FR seeds converge over training — opposite to the language pattern. Figures 4 and 5 provide the normalised perspective: by expressing FR–LR CKA as a fraction of the FR–FR baseline, the rank ordering becomes directly comparable across domains.

B Aim 2: Stabilisation Analysis

Figure 6 provides the full stabilisation analysis supporting §4.1. Panel (a) shows that higher rank stabilises earlier in both domains, consistent with the interpretation that higher-rank models traverse productive gradient directions more rapidly. Panel (b) shows that all language runs — including FR — stabilise by exhaustion, while vision LR_r128 stabilises by convergence. Panel (c) makes the init-CKA threshold visible directly: convergence runs sit above 0.65, exhaustion runs sit below. Panel (d) reveals the layer gradient in vision: shallow FR layers converge while deep layers exhaust, a pattern absent in language where all layers show uniform exhaustion.

C Aim 3: Supporting Gradient eRank Analyses

Figure 7 supplements the gradient starvation evidence in Figure 2. Panel (a) shows the ratio at the final checkpoint across all ranks: doubling rank approximately halves the ratio in both domains, but language ratios are substantially larger at every rank. Panel (b) separates structural eRank (determined by the rank constraint and largely fixed) from gradient eRank (dynamic, reflecting optimisation). The structural eRank of FR weight matrices decreases slightly over training (spectral consolidation), while gradient eRank is more variable — confirming that the two measures capture qualitatively different properties and that gradient eRank is the more informative diagnostic for the starvation claim.

D Aim 4: Spectral Intervention Details

Figure 8 provides the full Aim 4 O4.1 results: CKA and task-metric comparisons for all four intervention variants at t^* in both domains. Alpha-flattening is the most effective variant in both cases. The combined variant (alpha-flat + drift-scaled) consistently underperforms alpha-flat alone, suggesting that drift scaling adds noise rather than signal — spectral geometry rather than drift magnitude is the operative factor. Figure 9 shows

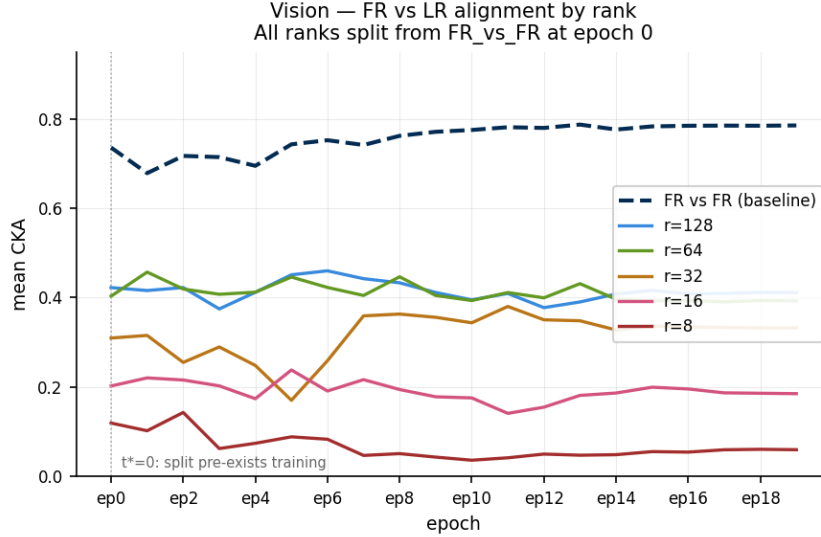


Figure 3: **Vision (CIFAR-10): FR-LR CKA trajectories by rank.** Cross-run CKA between FR and LR models at each rank across all 20 epochs. The split is present at epoch 0, before any gradient update. FR-FR CKA increases over training ($0.736 \rightarrow 0.785$; ratio $1.07\times$): two FR seeds converge in vision, opposite to the language pattern in Figure 1(a). Vision LR_{r128} shows a shallower divergence, consistent with its convergence-type stabilisation in Appendix B.

Domain	Variant	Mean CKA	CKA gain	Task metric	Task gain
Language	Baseline	0.082	—	489.7 PPL	—
Language	Drift-scaled	0.104	+27%	478.5 PPL	−2%
Language	Alpha-flat	0.318	+288%	188.2 PPL	−62%
Language	Combined	0.198	+141%	268.2 PPL	−45%
Vision	Baseline	0.318	—	0.223 Acc ⁺	—
Vision	Drift-scaled	0.362	+14%	0.234 Acc ⁺	+5%
Vision	Alpha-flat	0.516	+62%	0.253 Acc ⁺	+13.5pp
Vision	Combined	0.470	+48%	0.249 Acc ⁺	+12pp

Table 2: Spectral intervention results at t^* (Aim 4, O4.1). CKA gain and task gain are relative to the baseline (untreated LR) in each domain. Alpha-flattening is the most effective variant in both domains; the combined variant underperforms alpha-flat alone.

the performance–alignment scatter for both domains, making the dissociation geometrically visible: DSN conditions occupy the high-performance, low-alignment region regardless of domain or DSN mode.

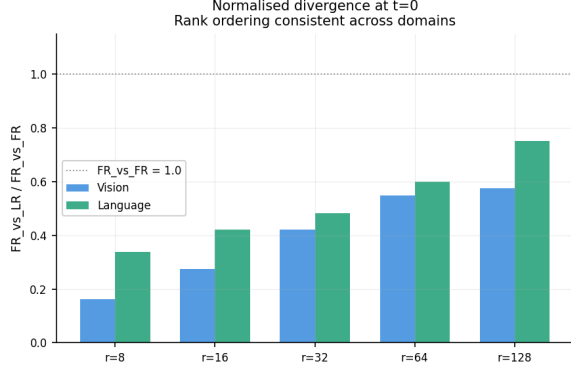
E Aims 5 and 6: Performance Gap and Recovery

Figure 10 shows the performance gap scaling in full. The key observation is that the gap decreases with rank at approximately the same rate as the gradient eRank ratio in Figure 2: doubling rank approximately halves both. This parallel scaling supports the view that gradient dimensionality is an important mediator of the performance cost, though parallel scaling does not establish a direct causal link.

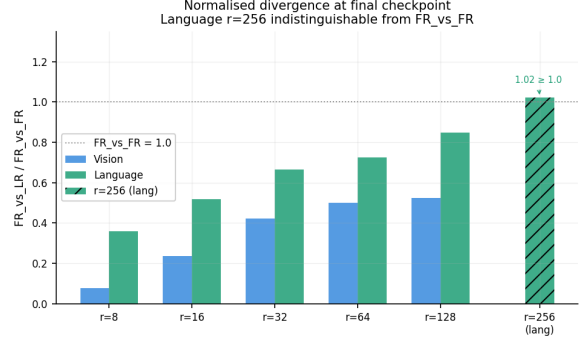
Figure 11 shows the full recoverability results. Panel (a) makes the contrast between language (monotonically increasing recovery) and vision (all-or-nothing recovery) directly visible. Panel (b) shows the residual gap under the most aggressive recovery condition in each domain, confirming that approximately half the gap remains unrecovered regardless of how many layers are reinitialised or how long fine-tuning continues.

F Sensitivity of the Stabilisation Threshold

The convergence/exhaustion classification in §2 uses two thresholds: $\tau_{\text{vel}} = 0.995$ for self-CKA velocity and $\tau_{\text{init}} = 0.65$ for init-CKA at stabilisation. Here we assess the sensitivity of the classification to these choices.



(a) Normalised divergence at $t=0$.



(b) Normalised divergence at final checkpoint.

Figure 4: **FR–LR CKA normalised by FR–FR baseline, both domains.** Values below 1 indicate that FR–LR divergence exceeds natural seed-level variability. Rank ordering is consistent and monotonic across domains at both time points. Language LR_r256 exceeds 1.0 at the final checkpoint (near-parity regime), while LR_r128 remains below 1.0 in both domains.

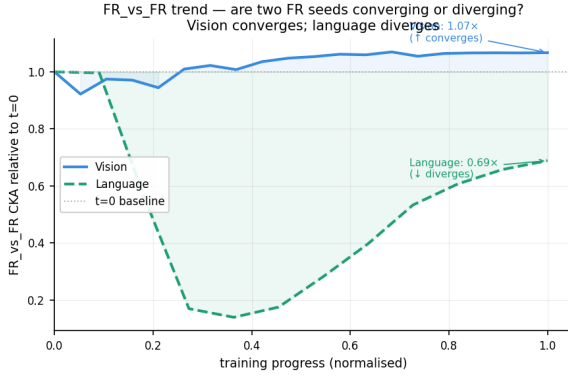


Figure 5: **FR–FR CKA over training, normalised to $t=0$ as 1.0.** Both series start at 1.0. Vision FR seeds converge ($1.07\times$ by the final epoch); language FR seeds diverge ($0.69\times$ by 50M tokens). Fill shading marks convergent (above baseline) and divergent (below baseline) regimes. This domain-level difference is independent of rank and reflects the structure of each domain’s optimisation landscape.

Velocity threshold τ_{vel} . Across $\tau_{\text{vel}} \in \{0.990, 0.995, 0.999\}$, the qualitative ordering of stabilisation points is preserved: higher rank stabilises earlier, and vision LR_r128 stabilises earliest. Absolute checkpoint values shift by at most 1–2 epochs (vision) or 5–10M tokens (language), but no run changes its relative position or mechanism classification. We adopt $\tau_{\text{vel}} = 0.995$ as it best separates runs with clearly stable representations from those still evolving.

Init-CKA threshold τ_{init} . At $\tau_{\text{init}} = 0.55$: all results remain identical to the main ones. At $\tau_{\text{init}} = 0.75$: vision LR_r64 (two layers) and FR vision (two deep layers) reclassify from conver-

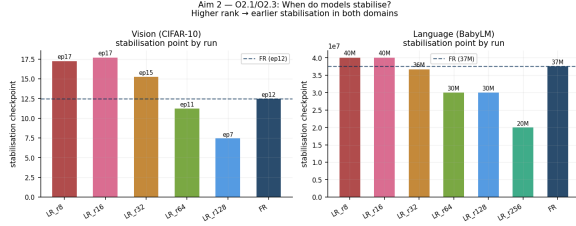
gence to exhaustion; all language runs remain at exhaustion. The primary claim — that all language runs stabilise by exhaustion including FR language — holds under all three threshold settings. The convergence classification for vision LR_r128 is somewhat sensitive at the margins and we note this as a qualification on that specific finding.

Alternative interpretation. Exhaustion could reflect a smaller effective step size at stabilisation rather than directional depletion. We assess this by comparing gradient eRank at t_s : vision FR shallow layers (convergence) show 62 ± 4 ; LR_r8 (exhaustion) show 11 ± 1 — a fivefold difference consistent with the directional interpretation. In language, all runs show similarly low gradient eRank at stabilisation, consistent with the universal exhaustion pattern. A formal decomposition is left to future work.

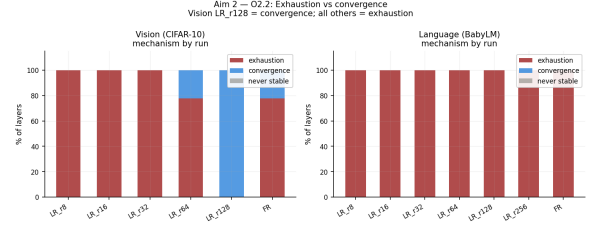
G Layer-wise CKA Analyses

Mean CKA is reported throughout the main paper. Here we confirm that key findings hold at the individual layer level.

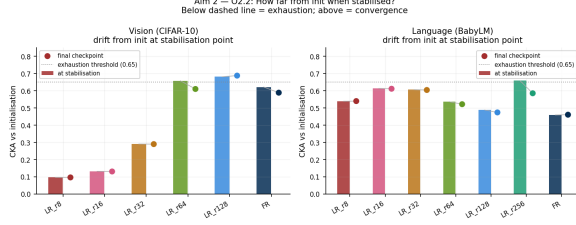
Vision. At epoch 0, all nine ViT layers show FR–LR CKA below the FR–FR baseline, confirming that the pre-existing split holds at every layer individually. Shallow layers (L0–L2) show systematically higher CKA than deep layers (L6–L8) throughout training, consistent with the convergence pattern in shallow FR layers. The layer ordering is stable — no layer crosses another — indicating that the relative difficulty of alignment is fixed early.



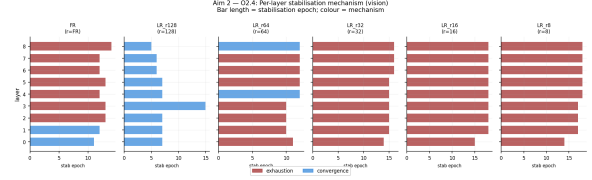
(a) Mean stabilisation checkpoint by run.



(b) % exhaustion / convergence per run.



(c) Init-CKA at t_s ; dashed line at threshold 0.65.



(d) Per-layer mechanism profile (vision).

Figure 6: **Stabilisation analysis, both domains (Aim 2).** (a) Higher rank stabilises earlier in both domains. (b) All language runs (including FR) stabilise by exhaustion; vision LR_{r128} stabilises by convergence. (c) Init-CKA at t_s separates the two regimes at threshold 0.65. (d) Vision layer profile: shallow FR layers converge; deep layers exhaust. No such gradient exists in language.

Language. All twelve GPT-2 layers are already below the FR–FR baseline at 8K tokens. Unlike vision, language layers show a near-uniform CKA profile across depth (spread < 0.01 at most checkpoints), consistent with the universal exhaustion pattern and the absence of a layer hierarchy. All layers show a CKA minimum at $\sim 524\text{K}$ – 1M tokens present in both FR–LR and FR–FR comparisons, suggesting it reflects the data distribution rather than the rank constraint. These layer-wise patterns support the use of mean CKA as the primary metric: The mean captures the dominant trend in both domains.

H Compute and Hyperparameter Details

Hardware. All experiments run on NVIDIA A100 80GB GPUs. Language rank sweeps (14 runs $\times$ 12 checkpoints) required approximately 280 A100 GPU-hours in total. Vision experiments (12 runs $\times$ 20 epochs) required approximately 48 A100 GPU-hours.

Language hyperparameters.

- Architecture: GPT-2, 12 layers, 12 heads, embedding dimensionality 768, context 128, $\sim 85\text{M}$ parameters (FR)
- Optimiser: AdamW, $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$, weight decay 0.01
- Learning rate: 10^{-4} , cosine decay, 500-step warm-up

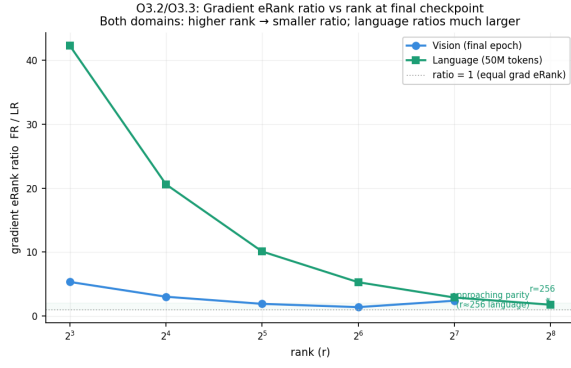
LR mult.	Init. scale mult.	Peak LR	Val. PPL
0.5	0.5	0.00015	218.07
0.5	1.0	0.00015	1860.69
0.5	2.0	0.00015	17531.11
1.0	0.5	0.0003	210.81
1.0	1.0	0.0003	641.46
1.0	2.0	0.0003	5158.22
2.0	0.5	0.0006	177.68
2.0	1.0	0.0006	532.03
2.0	2.0	0.0006	921.93
3.0	0.5	0.0009	176.13
3.0	1.0	0.0009	444.83
3.0	2.0	0.0009	688.60

Table 3: Full hyperparameter sweep grid for LR at $r = 8$. Learning rate and initialisation-scale multipliers are relative to the shared FR/LR baseline hyperparameters (Appendix H). The best configuration (bold) is retrained at the full training budget as LR-tuned (Section 3.4).

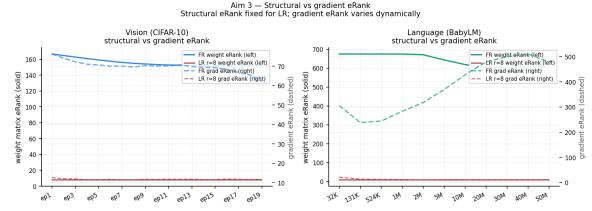
- Batch size: 64 sequences; token budget: 50M tokens
- Precision: BF16

Vision hyperparameters.

- Architecture: ViT, patch 4, embedding dimensionality 192, depth 9, heads 3, $\sim 4.3\text{M}$ parameters (FR)
- Optimiser: AdamW, $\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.05
- Learning rate: 10^{-3} , cosine decay, 5-epoch warm-up

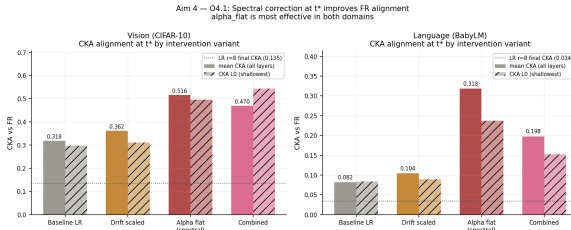


(a) Ratio at final checkpoint vs. rank ($\log_2$ scale).

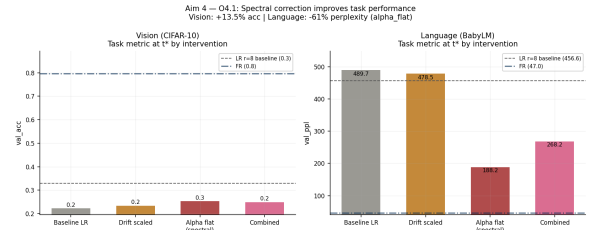


(b) Structural vs. gradient eRank over training.

Figure 7: **Supporting gradient eRank analyses (Aim 3).** (a) Doubling rank approximately halves the gradient eRank ratio at the final checkpoint in both domains. Language ratios are $\sim 8\times$ larger than vision at equivalent ranks. (b) Structural (weight matrix) eRank is largely fixed by the rank constraint; gradient eRank is dynamic and not determined by structural eRank. FR weight eRank decreases slightly over training (spectral consolidation); FR gradient eRank expands substantially in language but remains stable in vision.



(a) CKA at t^* by intervention variant.



(b) Task metric at t^* by intervention variant.

Figure 8: **Spectral correction at t^* , both domains (Aim 4, O4.1).** Alpha-flattening is the most effective variant: CKA +288% (language), +62% (vision); perplexity -62% ; accuracy $+13.5$ pp. The combined variant underperforms alpha-flat alone in both domains, indicating drift scaling adds noise rather than signal. Dashed lines show untreated LR baseline CKA.

- Batch size: 128 images; training: 20 epochs on CIFAR-10
- Precision: FP32

I Seed-Level Variance

We train two independently seeded runs per condition. Table 4 reports per-seed values for the headline metrics. Variance is consistently small relative to the effect sizes reported in Table 1: the FR–LR CKA split (0.046 vs. baseline 0.136) far exceeds the within-condition SD (≤ 0.005), and the gradient eRank ratio (42.3 for LR_r8) far exceeds any seed-level fluctuation (≤ 2.1). Qualitative trends — rank ordering, mechanism classification, ratio growth direction — are consistent across both seeds in all conditions.

Metric	Condition	Seed 1	Seed 2
FR–LR CKA (8K)	LR_r8 lang	0.044	0.048
FR–LR CKA (8K)	LR_r8 vis	0.081	0.087
FR–FR CKA (8K)	FR lang	0.132	0.140
FR–FR CKA (8K)	FR vis	0.723	0.748
eRank ratio (final)	LR_r8 lang	40.9	43.7
eRank ratio (final)	LR_r8 vis	5.1	5.5

Table 4: Per-seed values for headline metrics. Differences between seeds are small relative to FR–LR effect sizes.

J Probing Methodology: Mutual k -NN Alignment and BLiMP

J.1 Mutual k -Nearest-Neighbour Alignment

As an alignment metric independent of CKA, we adopt mutual k -NN alignment (Huh et al., 2024). Given activations $\mathbf{H}_a, \mathbf{H}_b \in \mathbb{R}^{n \times d}$ from the LR and FR models on the same n held-out inputs (Section 3.3), we ℓ_2 -normalise each row and compute the cosine similarity matrix within each represen-

Aim 4 — Dissociation: DSN improves task metric WITHOUT improving CKA
Better performance $\neq$ more FR-like representations

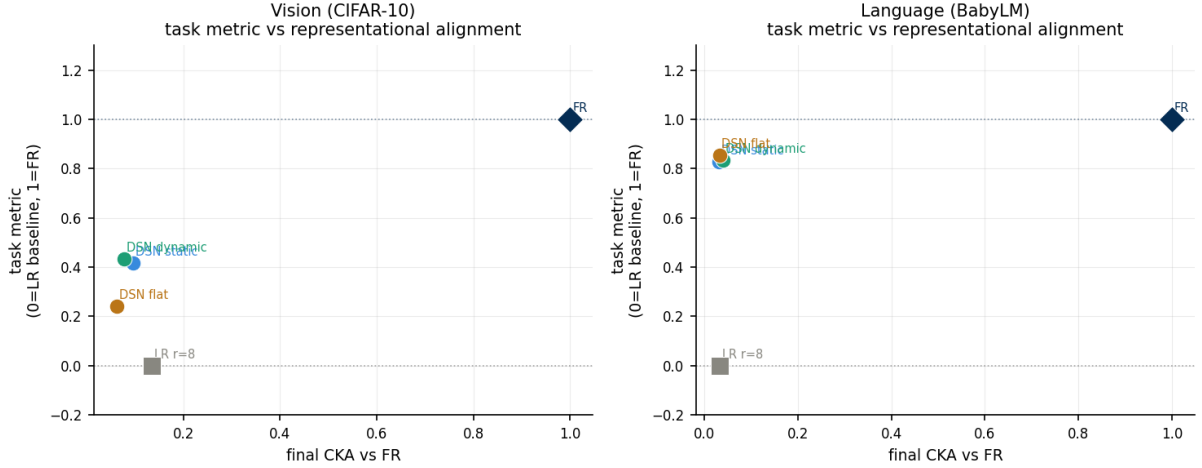


Figure 9: **Task performance vs. final CKA, both domains (Aim 4, O4.2).** Each point is a training configuration plotted by final CKA vs. FR (x -axis) and normalised task metric (y -axis; 0=LR baseline, 1=FR ceiling). DSN conditions (triangles) cluster in the high-performance, low-alignment region in both domains. FR (diamond) sits at the top right in each panel. The geometric separation confirms that closing the performance gap does not imply closing the representational gap, in both language and vision.

tation space. For each example i , we identify its k nearest neighbours (excluding itself) separately in $\mathbf{H}_a$ and $\mathbf{H}_b$, using $k = 10$. Alignment for example i is the fraction of these two neighbour sets that overlap:

$$\text{align}_i = \frac{|\mathcal{N}_a^k(i) \cap \mathcal{N}_b^k(i)|}{k}.$$

The reported score is the mean of align_i over all n examples, in $[0, 1]$. A score of 1 indicates that the two representation spaces agree perfectly on local neighbourhood structure; the expected score under no relationship between the spaces is $k/(n-1)$. Unlike CKA, which is sensitive to global linear-subspace correspondence, mutual k -NN alignment depends only on local neighbour ordering and is invariant to rotation and uniform rescaling of either representation space. We compute this metric per layer, on the same activations and held-out batch used for CKA, and report the mean across transformer layers.

As with CKA, we establish a natural baseline by computing mutual k -NN alignment between two independently-seeded FR runs. We note this baseline behaves differently under mutual k -NN alignment than under CKA: While the FR–FR CKA baseline decreases over training in language (Figure 5), the FR–FR mutual k -NN baseline increases substantially, more than doubling from initialisation to the final checkpoint. We report this discrep-

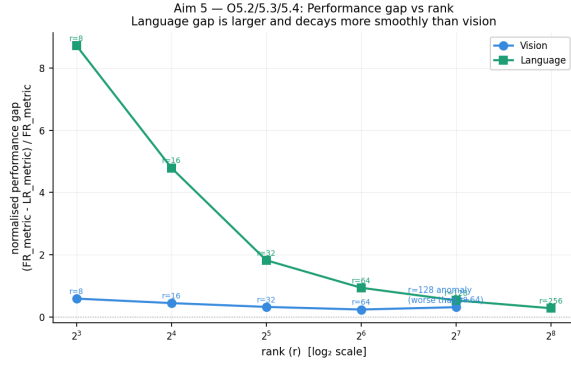
ancy directly rather than reconciling it; we discuss it as an open question in Section 5, and normalise all mutual k -NN results to their own baseline rather than to CKA’s.

J.2 BLiMP Functional Probing

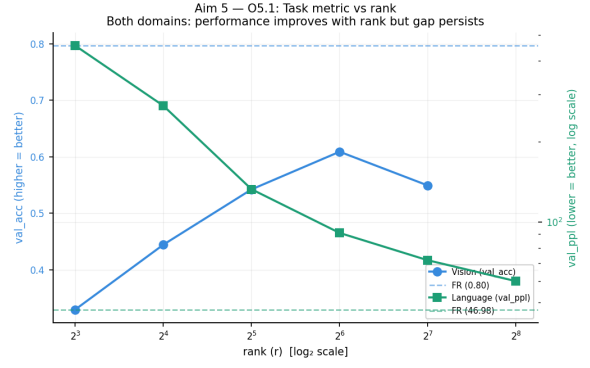
We assess functional, downstream competence using BLiMP (Warstadt et al., 2023b), a benchmark of minimal grammatical pairs, each consisting of a grammatical sentence and a minimally-perturbed ungrammatical counterpart isolating a single syntactic phenomenon. For each pair (s^+, s^-) , we compute the total sequence log-likelihood under teacher forcing,

$$\log p(s) = \sum_{t=1}^{|s|-1} \log p(s_{t+1} \mid s_{1:t}),$$

and score the pair correct if $\log p(s^+) > \log p(s^-)$. We evaluate 1,600 pairs per model, drawn from eight phenomena spanning agreement, negative polarity items, and wh-movement, and report overall accuracy together with a two-sided binomial test against chance ($p = 0.5$). For a subset of runs, we additionally cross-reference each checkpoint’s BLiMP-sample perplexity (mean negative log-likelihood over evaluated grammatical sentences) against the model’s own training-log validation perplexity at that checkpoint, as a sanity check that the evaluated checkpoint reflects a converged

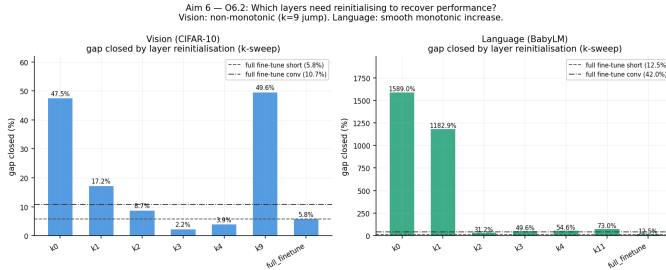


(a) Normalised gap δ_r vs. rank ($\log_2$ scale).

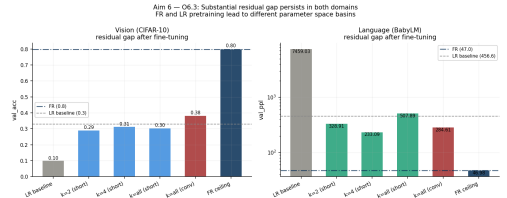


(b) Absolute task metric vs. rank.

Figure 10: Performance gap vs. rank, both domains (Aim 5). (a) Language gaps decrease monotonically and are $\sim 15\times$ larger than vision at equivalent ranks. Vision shows a non-monotonic anomaly at $r=128$, consistent with overfitting at higher capacity on CIFAR-10. Doubling rank approximately halves the gap in language, mirroring the gradient eRank ratio scaling in Figure 2. (b) Absolute task metrics confirm the pattern; FR reference lines (dashed) show the performance ceiling in each domain.



(a) Gap closed by k -sweep (short budget).



(b) Residual gap under key recovery conditions.

Figure 11: Recoverability analysis, both domains (Aim 6). (a) Language: recovery increases monotonically with k ; each reinitialised block contributes incrementally, consistent with the layer-uniform CKA profiles in Figure 1(a). Vision: all-or-nothing recovery; substantial improvement only at $k=9$ (all blocks), consistent with the layer-graded profiles in Appendix A. Dashed lines show full fine-tune performance under short and convergence budgets. (b) $\sim 51\%$ (vision) and $\sim 58\%$ (language) of the gap remains unrecovered under the most aggressive recovery condition.

rather than a mid-correction model state; results are unaffected in all reported runs.

K Aim 7: Functional Transfer via BLiMP and Mutual k -NN Alignment

K.1 BLiMP Accuracy Across Interventions

Table 5 reports BLiMP accuracy, final CKA relative to the natural FR–FR baseline, and validation perplexity for each of the eight interventions evaluated: LR_r8, the three DSN variants, GradTrunc, ReLoRA, LR-tuned, and spectrum-matched LR_r8.

Notably, CKA and perplexity do not always agree on relative ordering. The three DSN variants achieve some of the best perplexity of any LR-family condition while remaining near chance on BLiMP, and GradTrunc, with the best perplex-

Run	CKA		
	BLiMP	(% baseline)	Val. PPL
GradTrunc	0.578	68.6%	72.8
ReLoRA_r8	0.495	56.1%	604.6
DSN-dynamic	0.514	42.6%	114.7
LR-tuned	0.520	42.3%	138.2
LR_r8 (baseline)	0.556	37.2%	456.6
DSN-flat	0.488	34.8%	109.7
DSN-static	0.515	33.3%	117.2
Spectrum-matched	0.533	30.5%	438.7

Table 5: BLiMP accuracy, final CKA as a percentage of the natural FR–FR baseline, and validation perplexity for each intervention. Per-run significance against chance is reported in Table 6.

ity overall, also shows the highest BLiMP accuracy. Across all eight runs, CKA correlates with BLiMP accuracy more strongly than perplexity does ($r=0.40$ vs. $r=-0.08$); CKA and perplexity

are themselves uncorrelated ($r = -0.05$), consistent with the dissociation reported under DSN in §4.3.

Table 6 reports a two-sided binomial test against chance ($p = 0.5$, $n = 1,600$ pairs) for each run individually.

Run	BLiMP acc.	Distinguishable from chance?
GradTrunc	0.578	yes
LR_r8 (baseline)	0.556	yes
Spectrum-matched	0.533	yes
LR-tuned	0.520	no
DSN-static	0.515	no
DSN-dynamic	0.514	no
DSN-flat	0.488	no
ReLoRA_r8	0.495	no

Table 6: Binomial test against chance for each run, $n = 1,600$ pairs.

K.2 Mutual k -NN Alignment

We compute mutual k -NN alignment ($k=10$, Appendix J) for a subset of runs as an independent check on the CKA-based representational gap. Table 7 reports final-checkpoint alignment relative to the natural FR–FR baseline under this metric.

Run	Mutual k -NN align.	% of FR–FR baseline
FR–FR baseline	0.702	100.0%
GradTrunc	0.528	75.2%
DSN-static	0.366	52.1%
LR_r8	0.061	8.7%

Table 7: Mutual k -NN alignment, final checkpoint, relative to the natural FR–FR baseline under this metric.

The representational gap between LR-family models and FR persists under mutual k -NN alignment, confirming it is not an artefact specific to CKA’s particular notion of similarity. We note two caveats. First, the FR–FR baseline itself behaves differently under the two metrics (Table 8): it declines over training under CKA, but rises substantially, more than doubling, under mutual k -NN alignment.

We do not have a full explanation for this reversal and flag it as an open question. Second, the two metrics do not always agree on relative ordering between runs: mutual k -NN alignment ranks DSN-static above LR_r8, the reverse of both CKA’s ordering and what BLiMP accuracy in Table 5 supports. We report both metrics rather than the one more favourable to our claims; raw magnitudes are not comparable across metrics, and only each

Tokens	FR–FR CKA	FR–FR mutual k -NN
0	0.136	0.318
524K	0.020	0.382
5M	0.038	0.627
20M	0.072	0.682
50M	0.094	0.702

Table 8: FR–FR baseline trend under CKA versus mutual k -NN alignment. CKA declines over training; mutual k -NN alignment rises substantially.

metric’s own baseline-normalised trend should be compared across runs.